



Machine Learning for Quantum Phase Transitions

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Abstract

Quantum many-body systems exhibit rich emergent behavior, with quantum phase transitions (QPTs) providing key insights into strongly correlated matter. The Bose-Hubbard (BH) model serves as a paradigmatic framework to study such transitions, capturing the interplay between kinetic energy and on-site interactions that give rise to the superfluid-Mott insulator transition. However, accurate characterization of QPTs becomes computationally demanding as system sizes grow, motivating the integration of machine learning (ML) techniques into quantum many-body research.

In this work, we explore the use of neural networks to classify quantum phases and detect critical points in the Bose-Hubbard model. By training supervised & unsupervised learning models on observables such as superfluid order parameter, and local occupation number we demonstrate that neural architectures can learn nontrivial order parameters and capture universal features of QPTs without prior knowledge of the underlying physics. This study highlights the potential of approaches like machine learning (ML), pointing towards data-driven strategies for uncovering novel phases of matter and advancing the theoretical understanding of quantum criticality [1-5].

Introduction

The Bose-Hubbard model is given by the following equation:

$$H = -J \sum_i (a_i^\dagger a_{i+1} + H.c.) + \frac{U}{2} \sum_i n_i(n_i - 1) - \mu \sum_i n_i \dots (1)$$

Here i is the site index and,

- $a_i^\dagger(a_i)$ $\rightarrow$ Bosonic creation / annihilation operator
- n_i $\rightarrow$ on-site occupation number operator
- μ $\rightarrow$ chemical potential
- J $\rightarrow$ hopping amplitude
- U $\rightarrow$ on-site interaction

We can obtain the phase diagram for this model using a method like mean-field (MF)[1], Quantum Monte Carlo (QMC), or Density Matrix Renormalization Group (DMRG). Under single-site MF decoupling approximation the model transforms to:

$$H_i^{MF} = -\phi(a_i^\dagger + a_i) + \phi^2 + \frac{\bar{U}}{2} n_i(n_i - 1) - \bar{\mu} n_i \dots (2)$$

- $\bar{U} = U/J, \bar{\mu} = \mu/J$ $\rightarrow$ scaled parameters
- ϕ $\rightarrow$ superfluid order parameter

The above Hamiltonian can be solved self-consistently to obtain the complete phase diagram [1]. Upon solving this model gives two phases, namely superfluid (SF) at non-integer densities and Mott-insulator (MI) at integer densities, subject to the conditions:

- $U/J \gg 1 \Rightarrow$ Mott-insulator, $\phi = 0, n \in \mathbb{Z}$
- $U/J \ll 1 \Rightarrow$ superfluid, $\phi \neq 0$

The phase diagram obtained using Eq.(2) is given in Fig.1. The red (blue) area represents the SF (MI) phase. We use the data corresponding to Fig.(1) for training our models as well.

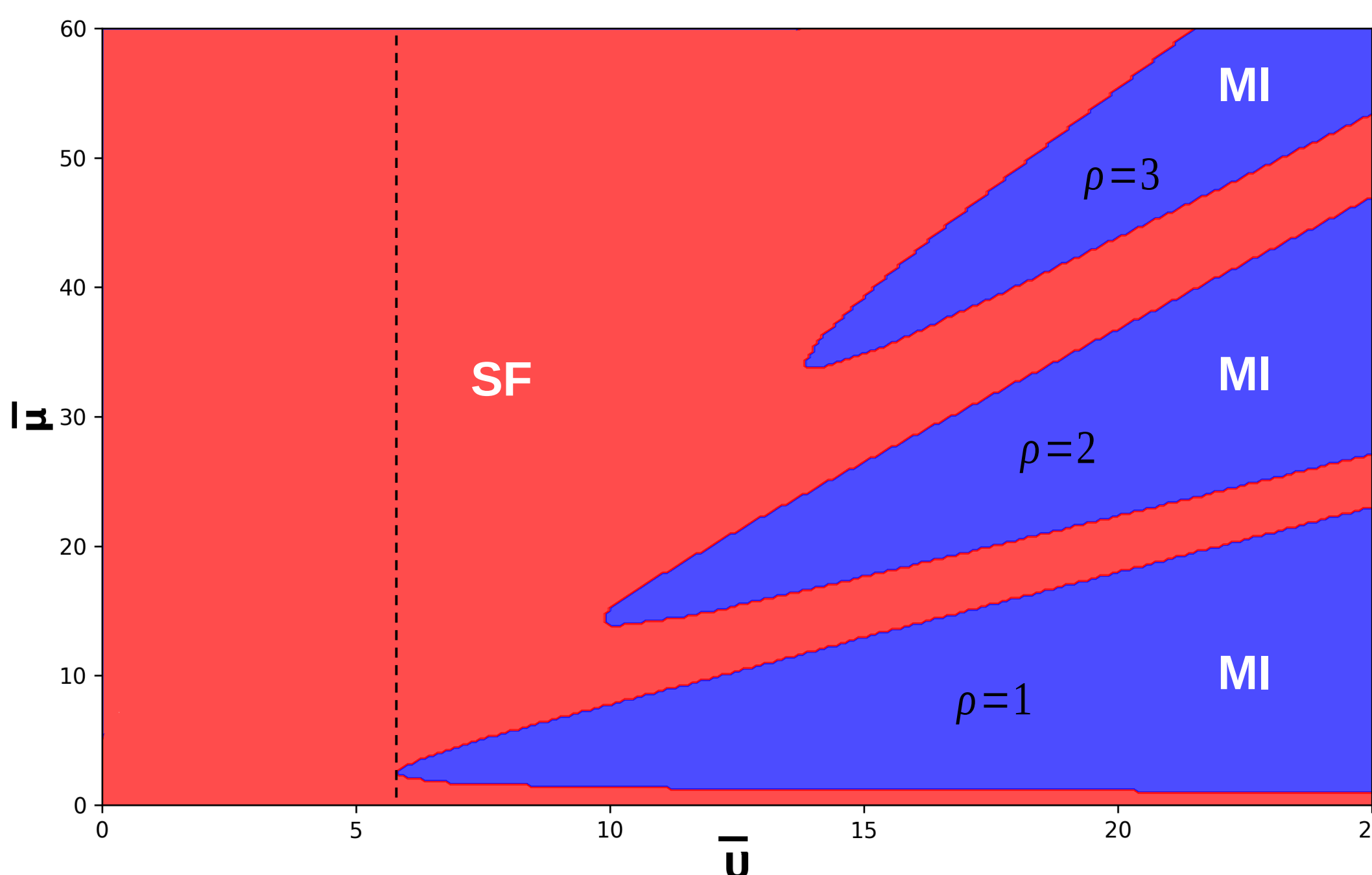


Fig.1 : Single-site MF phase diagram. The numbers inside the lobes indicate the density of the MI phases. The dashed line indicates the position of SF-MI(1) critical point (~ 5.83).

Methodology

We use the following four ML models in combination with the data obtained, as mentioned earlier :

- (i) **Multi-Layer Perceptron (MLP)**,
- (ii) **Keras-based Neural Network**,
- (iii) **Physics Informed Neural Networks + Keras**, and
- (iv) **PINN + Bayesian Surrogate Model**.

In the following we briefly describe the working of each of the above:

(i) Multi-Layer Perceptron (MLP) : The MLP serves as the baseline data-driven model, trained on BH parameters (U, μ) to classify quantum phases. It employs two hidden layers (50, 50) with ReLU activation and the Adam optimizer (LR = 0.001). It is a generic feed-forward NN network without domain specific constraints and is moderately reliable

(ii) Keras-based Neural Network : This model extends the baseline MLP framework by introducing additional flexibility through Keras' modular architecture — enabling custom callbacks, dropout regularization, and adaptive early stopping. It's trained on normalized BH parameters (U, μ) using ReLU activations and the Adam optimizer (LR = 0.001). Unlike the MLP, the Keras NN offers a flexible base for adding physics-informed constraints or surrogate models later.

(iii) Physics Informed Neural Networks (PINN)+ Keras : This model combines a feed-forward NN with physics-informed constraints via a pre-trained Keras surrogate to classify phases and predict ϕ^2 and ρ . It uses two hidden layers (128, 64, tanh) and a composite loss: binary cross-entropy (phase), MSE (ϕ^2, ρ), and a physics residual enforcing surrogate consistency. The physics residual is dynamically weighted using a ramped parameter λ_{phys} , guiding the network to honor underlying physical laws while fitting data. Trained with: Adam (LR = 0.001), 200 epochs.

(iv) PINN enhanced with a Bayesian surrogate : With parameters (U, μ) as inputs, model processes them through two hidden layers (128, 64), with $\tanh$ activations and outputs phase (sigmoid), ϕ^2 (ReLU), ρ (ReLU), and a physics residual enforcing consistency with the Bayesian surrogate. The loss combines binary cross-entropy for phase, MSE for ϕ^2 and ρ , and the surrogate-based physics residual, with a ramped physics weight to stabilize training. Monte Carlo dropout in the surrogate provides uncertainty estimates, identifying ambiguous regions. Training is same as before.

Results

(i) MLP : The model achieved high accuracy (>98%) across all dataset scales and successfully captured the MI-SF transition boundary. However, its predictive reliability diminished outside the training domain.

(ii) Keras based NN: The model could effectively learn the nonlinear phase boundaries as well. It also achieved strong generalization within the training domain.

(iii) PINN + Keras: This approach achieved even higher accuracy (>99% on full data) and robustly captures phase boundaries, even in sparsely sampled or boundary regions.

(iv) PINN + Bayesian surrogate method: The model achieves 99.3% accuracy on the full dataset and remarkably maintained 96–98% accuracy even on the reduced datasets (4k–30k points), capturing phase boundaries reliably. The integrated physics ensures physically consistent predictions, while Bayesian uncertainty quantifies confidence, making the approach robust, interpretable, and data-efficient.

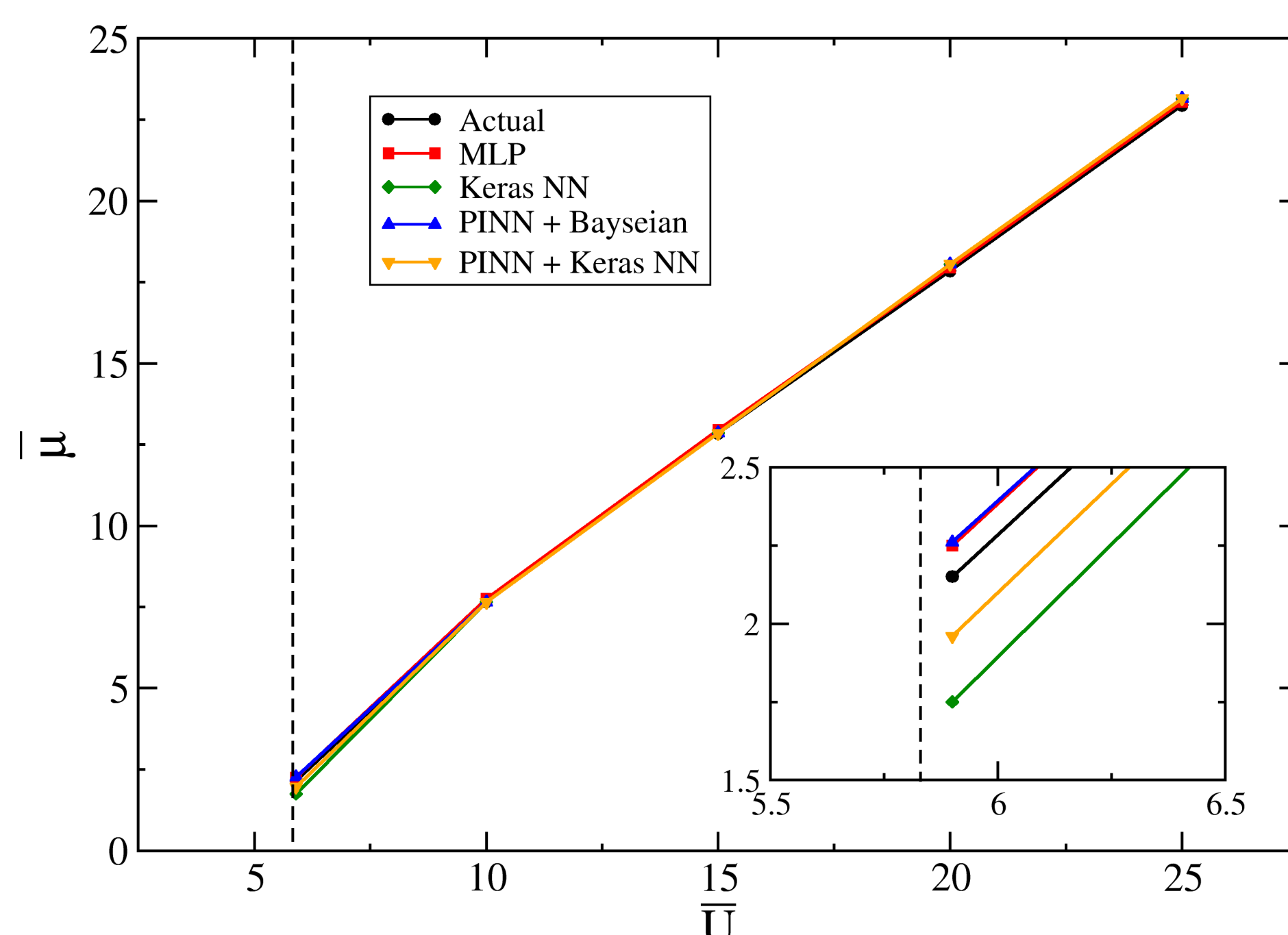


Fig.2 : Comparison of actual and ML predicted SF-MI(1) upper boundary. Inset shows a zoomed in part close to the tip of MI(1) lobe. Critical point (U/J) is at ~ 5.83 (dashed line).

U	Actual μ	Predicted μ			
		MLP	Keras-NN	PINN + Keras NN	PINN + Bayesian
10	7.65	7.75	7.65	7.65	7.65
15	12.85	12.95	12.85	12.85	12.85
20	17.85	17.95	18.05	18.05	18.05
25	22.95	23.05	23.15	23.15	23.15

Table 1 : Actual vs. predicted μ for MI(1) upper boundary.

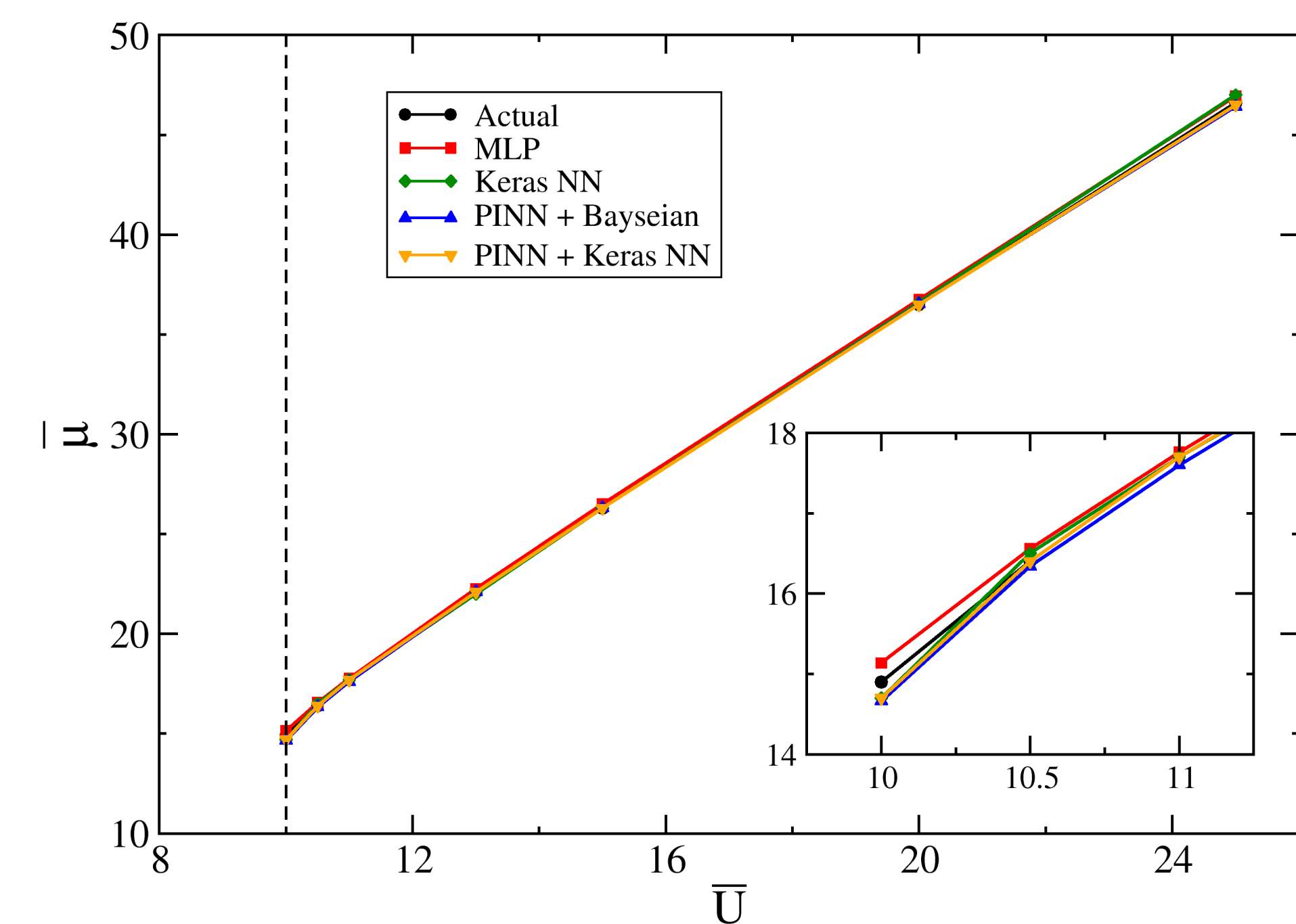


Fig.3 : Comparison of actual and ML predicted SF-MI(2) upper boundary. Inset : Zoomed in part close to the tip of MI(2) lobe.

U	Actual μ	Predicted μ			
		MLP	Keras-NN	PINN + Keras NN	PINN + Bayesian
10	14.9	15.1	14.7	14.7	14.66
15	26.3	26.3	26.3	26.3	26.30
20	36.5	36.7	36.6	36.5	36.52
25	46.6	46.9	47.0	46.5	46.42

Table 1 : Actual vs. predicted μ for MI(1) upper boundary.

Comparison of various metrics of the methods we have used:

Full data set : [163401 rows x 8 columns]

Model	Accuracy	Precision	Recall	F1-score
MLP	0.9924	0.9933	0.991	0.9921
Keras-based NN	0.99	0.9897	0.9896	0.9896
PINN + Keras NN	0.9988	0.9986	0.9989	0.9988
PINN + Bayesian	0.9933	0.9929	0.9933	0.9931

Data set 1: [30684 rows x 8 columns]

Model	Accuracy	Precision	Recall	F1-score
MLP	0.9837	0.9837	0.9839	0.9837
Keras-based NN	0.9812	0.9814	0.9811	0.9812
PINN + Keras NN	0.9909	0.9909	0.9911	0.9909
PINN + Bayesian	0.9879	0.9889	0.9864	0.9876

Data set 2: [10684 rows x 8 columns]

Model	Accuracy	Precision	Recall	F1-score
MLP	0.9805	0.9815	0.9796	0.9804
Keras-based NN	0.9528	0.9528	0.9525	0.9526
PINN + Keras NN	0.9613	0.9608	0.9626	0.9612
PINN + Bayesian	0.9767	0.9793	0.9737	0.9761

Data set 3: [4684 rows x 8 columns]

Model	Accuracy	Precision	Recall	F1-score
MLP	0.9693	0.9745	0.964	0.9683
Keras-based NN	0.9146	0.9189	0.9071	0.9116
PINN + Keras NN	0.9441	0.9411	0.9468	0.9433
PINN + Bayesian	0.9667	0.9692	0.9643	0.9663

Conclusion

Machine learning provides an efficient and scalable approach to study QPTs in many-body systems such as the BHM. Conventional numerical methods like DMRG or QMC are accurate but computationally intensive, whereas ML models can classify phases and identify critical points with ease and comparable precision. Among the models explored, the PINN combined with a Bayesian surrogate achieved the highest accuracy ($\approx 99.3\%$) and demonstrated robustness even with limited data, while also providing uncertainty quantification. This hybrid physics-ML approach not only accelerates phase diagram generation but also enhances interpretability and reliability—paving the way for data-driven discovery of quantum phenomena and broader applications in computational condensed matter physics.

References

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